

Examen simulado ecuaciones diferenciales

Resolver

1.

$$e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y}$$

2.

$$y \ln(x) \frac{dx}{dy} = \left(\frac{y+1}{x} \right)^2$$

3.

$$\sin(3x)dx + 2y \cos^3(3x)dy = 0$$

4.

$$\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$$

5.

$$y' + 2xy = x^3$$

6.

$$y' = 2y + x^2 + 5$$

7.

$$(1+x) \frac{dy}{dx} - xy = x + x^2$$

8.

$$\cos x \frac{dy}{dx} + (\sin x)y = 1$$

9.

$$\cos^2(x)\sin(x) \frac{dy}{dx} + (\cos^3 x)y = 1$$

10.

$$(x^2 - 1) \frac{dy}{dx} + 2y = (x + 1)^2$$

11.

$$\left(2y - \frac{1}{x} + \cos(3x)\right) \frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y\sin(3x) = 0$$

12.

$$x \frac{dy}{dx} = 2xe^x - y + 6x^2$$

13.

$$\begin{aligned} (2y\sin x \cos x - y + 2y^2 e^{xy^2})dx \\ + (-x + \sin^2 x + 4xy^{xy^2})dy = 0 \end{aligned}$$

14.

$$(y^2 \cos x - 3x^2 y - 2x)dx + (2y \sin x - x^3 + \ln y)dy = 0$$

15.

$$y(x + y + 1)dx + (x + 2y)dy = 0$$

16

$$\cos x dx + \left(1 + \frac{2}{y}\right) \sin x dy = 0$$

17.

$$(y^2 + xy^3)dx + (5y^2 - xy + y^3 \sin y)dy = 0$$

18.

$$x dx + (x^2 y + 4y)dy = 0$$

Final boss

19.

$$(x^2 + y^2 - 5)dx = (y + xy)dy, \quad y(0) = 1$$

20.

$$x dx + y dy + (x^2 + y^2)x^2 dx = 0$$

Extra

21.

$$y dx + x dy + 4x^3 dx = 0$$